EPJ manuscript No.
(will be inserted by the editor)

Spectral Criticality from Global Monopoles to Plebański–Hacyan Spacetime

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the date of receipt and acceptance should be inserted later

Abstract. We investigate the spectral properties of a massless scalar field propagating on a one-parameter family of isotropic ultrastatic backgrounds in $(n+1)$ dimensions. Starting from the Einstein field equations, we construct a geometric sector that includes the Einstein static universe. The power-law solutions, characterized by the metric potential $\Phi(r) = ar^b$, exhibit a conical structure that, in three spatial dimensions, corresponds to the exterior geometry of a global monopole, with angular scale factor $\alpha = |1 + b/2|$. The exceptional configuration $b = -2$ corresponds to a nonsingular cylindrical geometry, identified in four spacetime dimensions with the Plebański–Hacyan spacetime.
We analyze the radial Klein–Gordon operator through its Sturm–Liouville formulation. For $b > -2$, the adopted regular Bessel modes and Dirichlet boundary conditions at a spherical cavity yield a discrete radial spectrum. We identify $b = -2$ as a distinguished spectral critical point, where the effective radial interval becomes unbounded and the Liouville coordinate acquires a logarithmic dependence. At this configuration, the inverse-square Schrödinger potential becomes constant and the discrete Bessel spectrum gives way to a continuous spectrum of plane wave modes. This transition is accompanied by the emergence of a cylindrical geometry with constant curvature invariants.
Our results establish a connection between ultrastatic geometries, global monopole exteriors, and the Plebański–Hacyan spacetime through the spectral structure of the massless Klein–Gordon operator.

PACS. XX.XX.XX No PACS code given

Introduction

In quantum field theory in curved backgrounds (QFTCB), the spectrum of quantum fluctuations is influenced by the underlying spacetime geometry, topology and boundary conditions, giving rise to vacuum polarization and Casimir phenomena. Highly symmetric backgrounds, such as the Einstein static universe and closed Friedmann models, have been extensively investigated in this context [1, 2]. In particular, ultrastatic spacetimes provide a useful framework for studying geometry-induced vacuum effects, since the dynamics of quantum fields can be reduced to a spectral problem on the spatial geometry [3,4].

A physically relevant example is provided by global monopoles, which arise as topological defects associated with spontaneous symmetry breaking. Their approximate exterior gravitational field is characterized by a solid angle deficit, leading to a conical spatial geometry [5]. The propagation of quantum fields on such backgrounds is particularly interesting because the idealized metric contains a naked singularity at the origin. Consequently, the spatial Klein–Gordon operator may admit inequivalent self-adjoint extensions, corresponding to different boundary conditions at the singular endpoint. As demonstrated in Ref. [6], these choices can modify the vacuum fluctuations of a massless scalar field, with the Dirichlet prescription recovering the expected Minkowski limit.

More generally, the determination of vacuum observables requires specifying the spectral properties of the underlying differential operator. In the canonical quantization approach [7], the Wightman two-point function can be constructed from a complete set of normalized field modes. The associated vacuum energy is formally obtained through a sum over eigenfrequencies, which generally requires regularization. Spectral zeta-function techniques provide a systematic framework for treating these divergent mode sums and their associated one-loop quantities [8–11].

In previous work [12], dimension-dependent effective geometries were investigated within the same ultrastatic framework. In particular, the influence of stress anisotropy and dimensionality on gravitational lensing was studied. Even though some geometries leave a characteristic coni-

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cal lensing, they were not proven to represent monopoles. Motivated by these considerations, in this work we propose an alternative power-law metric potential $\Phi(r) = ar^b$ and investigate a one-parameter family of isotropic ultrastatic geometries in $(n+1)$ dimensions. A distinguished member corresponds to the Plebański-Hacyan universe [13, 14]. The general ultrastatic sector also contains the Einstein static universe as a particular case. Furthermore, in three spatial dimensions, the power-law sector approximates to the exterior geometry of a global monopole. For $b \neq -2$, the spatial metric can be expressed in conical form, with angular parameter $\alpha^2 = (1+b/2)^2$. The conventional global monopole exterior is recovered with the deficit solid angle $\alpha^2 = 1 - 8\pi G_{\mathsf{N}}\eta^2$, where η is the scale of symmetry breaking.

The correspondence then provides a motivation to study vacuum fluctuations in the presence of a global monopole. We construct the massless Wightman two-point function through a separation of variables and the associated Sturm–Liouville formulation. For the power-law backgrounds with $b \neq -2$, the radial equation admits Bessel solutions. From this, we select the Dirichlet boundary condition. The choice is preferred since in this case the nontrivial interaction between the naked singularity and the quantum field is not probed by vacuum fluctuations and they coincide with the Minkowski limit (See Ref. [6]), as $\eta \to 0$ (or equivalently $b \to 0$). Our objective is then to investigate how the geometric deformation parameter b modifies the spectrum of the Klein–Gordon operator.

In Section 1 we derive a reduced field equation and analyze its exact solution branches. In Section 2 we quantize a massless scalar field propagating in these backgrounds. Finally, in Section 3 we show that the cylindrical Plebański-Hacyan solution defines a distinguished spectral critical point of the Klein–Gordon operator.

1 Geometry and Matter Content

In this section, we construct a class of ultrastatic, spherically symmetric geometries in $(n+1)$ dimensions and analyze the matter content required to support them. Higher-dimensional spacetimes have been extensively studied in classical gravity and provide a broad setting for investigations in semiclassical gravity [15]. We show that the Einstein field equations reduce to a single nonlinear, radial and second order differential equation for the metric potential, enabling a systematic characterization of stress-supported configurations. In particular, the power-law family exhibits a conical structure and, depending on the parameters, naked singularities. These features make the resulting background a suitable setting for studying quantum fields in the presence of nontrivial topological defects, such as global monopoles, while retaining analytical control over the spectral decomposition.

1.1 Generalized Einstein-universe sector

We consider a generalized Einstein-universe sector described by the isotropic ultrastatic metric

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \Phi(r)\,\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j\,, \tag{1}$$

in $n+1$ spacetime dimensions. This class contains the Einstein static universe as a particular member when expressed in stereographic coordinates and extends it to a broader family of anisotropic stress-supported geometries.

Let us write the Einstein field equations in $(1+n)$ spacetime dimensions as

$$G_{\mu\nu} = \kappa_n^2\, T_{\mu\nu}\,, \tag{2}$$

where $\kappa_n^2 = (8\pi G_{\mathsf{N}}^{(n)})^{-1}$ denotes the inverse gravitational coupling in $(n+1)$ spacetime dimensions, in natural units $c = \hbar = 1$. In conventional cgs units, the corresponding Newtonian gravitational constant has dimensions

$$[G_{\mathsf{N}}^{(n)}] = \mathrm{cm}^n\,\mathrm{s}^{-2}\,\mathrm{g}^{-1}\,. \tag{3}$$

For four-dimensional spacetime ($n = 3$), we recover $\kappa_3^2 = M_{\mathsf{P}}^2$, where $M_{\mathsf{P}}^2 = (8\pi G_{\mathsf{N}})^{-1}$ is the squared reduced Planck mass.

From Eqs. (79) of the Appendix 5, the Einstein tensor components are

$$G_{00} = \frac{1}{2}R\,, \tag{4a}$$

$$G_{0i} = 0\,, \tag{4b}$$

$$G_{ij} = \left(\mathcal{A} - \frac{1}{2}\Phi R\right)\delta_{ij} + \mathcal{B}\frac{x^i x^j}{r^2}\,. \tag{4c}$$

The structure of the Einstein tensor suggests that the corresponding matter source must include anisotropic stress. We therefore consider the most general static stress–energy tensor compatible with spherical symmetry, decomposed into energy density, isotropic pressure and a traceless anisotropic component.

Then, the Einstein field equations are satisfied by a static anisotropic stress–energy tensor of the form

$$T_{\mu\nu} = \rho(r)\,u_\mu u_\nu + P(r)\,h_{\mu\nu} + \tau_{\mu\nu}(r)\,, \tag{5}$$

where $u^\mu = \delta^\mu_0$ is the static unit timelike vector,

$$h_{\mu\nu} := g_{\mu\nu} + u_\mu u_\nu\,, \tag{6}$$

is the spatial projector and $\tau_{\mu\nu}$ denotes a traceless anisotropic stress tensor. The non–vanishing components are

$$\rho(r) = \frac{\kappa_n^2}{2}\,R\,, \tag{7a}$$

$$P(r) = \left(\frac{2-n}{n}\right)\frac{\kappa_n^2}{2}\,R\,, \tag{7b}$$

$$\tau_{ij}(r) = \Pi(r)\left(\frac{x^i x^j}{r^2} - \frac{1}{n}\delta_{ij}\right), \quad \Pi(r) = \kappa_n^2\mathcal{B}\,, \tag{7c}$$

$$\tau_{0\mu} = 0\,. \tag{7d}$$

Furthermore, from Eq. (7b) we see that the pressure is proportional to the energy density. Thus, the conservation law, $\boldsymbol{\nabla}_\mu T^{\mu k} = 0$, yields the radial hydrostatic equation

$$P' + \frac{\Pi'}{\Phi} + \left[\frac{n-1}{r\Phi} + \frac{(n-3)\Phi'}{2\Phi^2}\right]\Pi = 0\,, \tag{8a}$$

$$P(r) = -\left(\frac{n-2}{n}\right)\rho(r)\,, \tag{8b}$$

which provides a consistency relation between the anisotropic stress $\Pi(r)$, the metric potential $\Phi(r)$ and the energy density $\rho(r)$.

Reduced field equation

In the trivial case of $n = 1$, both the energy density and the isotropic pressure vanish. Moreover, the anisotropic stress $\Pi(r)$ also vanishes identically since directional asymmetries cannot be sustained in a single spatial dimension, reducing the scenario to a trivial vacuum.

For $n > 1$, the system of Eqs. (7) reduces to a non–linear second order differential equation for the metric potential, sourced by the energy density profile. We refer to this as the *reduced field equation*,

$$\frac{\Phi''}{\Phi} + \frac{n-1}{r}\frac{\Phi'}{\Phi} + \left(\frac{n-6}{4}\right)\left(\frac{\Phi'}{\Phi}\right)^2 + f(r)\Phi(r) = 0\,, \tag{9a}$$

$$f(r) := \frac{2\,\rho(r)}{(n-1)\,\kappa_n^2}\,. \tag{9b}$$

Consequently, once the reduced equation is solved for $\Phi(r)$, the matter variables follow from Eqs. (7), while Eq. (8) is automatically satisfied as a consequence of the conservation law. Notably, because of the nonlinearity, a given energy density profile does not determine a unique solution for the metric potential and multiple solution branches may arise.

1.2 Uniform energy density sector

Metric potential for the Einstein universe

To transform the spatial geometry of the Einstein static universe from standard hyperspherical to stereographic coordinates, we apply a stereographic projection. Starting with the spatial metric on an n-sphere of radius a,

$$\mathrm{d}l^2 = a^2(\mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega_{n-1}^2)\,, \tag{10}$$

we introduce the stereographic radial coordinate $r = a\tan(\chi/2)$.

Substituting the corresponding differential and identity

$$\mathrm{d}\chi = \frac{2a}{a^2+r^2}\,\mathrm{d}r\,, \quad \sin\chi = \frac{2ar}{a^2+r^2}\,, \tag{11}$$

into the line element yields

$$\mathrm{d}l^2 = \frac{4a^4}{(a^2+r^2)^2}(\mathrm{d}r^2 + r^2\mathrm{d}\Omega_{n-1}^2)\,. \tag{12}$$

Recognizing that the term in parentheses is simply the flat Euclidean metric written in spherical coordinates, i.e.

$$\begin{aligned}\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j &= \mathrm{d}r^2 + r^2\mathrm{d}\Omega_{n-1}^2\,,\\ \mathrm{d}\Omega_{n-1}^2 = \delta_{ij}\mathrm{d}n^i\mathrm{d}n^j\,, &\quad n^i := \frac{x^i}{r}\,,\end{aligned} \tag{13}$$

the full spacetime metric elegantly takes the manifestly conformally flat, isotropic form

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{4a^4}{(a^2+r^2)^2}\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j\,. \tag{14}$$

Einstein static universe

To connect our ultrastatic sector to the previous metric (14) first consider a uniform and positive energy density,

$$\rho(r) = \rho_0 > 0\,, \tag{15a}$$

$$A := \frac{2\rho_0}{(n-1)\kappa_n^2}\,. \tag{15b}$$

Motivated by the stereographic form of the Einstein static universe, consider the ansatz

$$\Phi(r) = (a + r^\gamma)^\beta\,. \tag{16}$$

Then, the reduced field equation admits the unique nontrivial solution

$$\gamma = 2\,, \qquad \beta = -2\,, \quad A = 4an\,. \tag{17}$$

Choosing the overall normalization to recover the standard stereographic parametrization yields

$${}^{(\mathrm{Ein})}\Phi(r) = \frac{4a^4}{(a^2+r^2)^2}\,, \tag{18}$$

which coincides with the metric potential of the Einstein static universe in isotropic stereographic coordinates.

From the Ricci scalar, Eq. (23), this geometry satisfies the Einstein field equations with

$${}^{(\mathrm{Ein})}\rho = \frac{\kappa_n^2}{2}\frac{n(n-1)}{a^2}, \tag{19a}$$

$${}^{(\mathrm{Ein})}P = -\frac{\kappa_n^2}{2}\frac{(n-1)(n-2)}{a^2}, \tag{19b}$$

$${}^{(\mathrm{Ein})}\Pi = 0, \tag{19c}$$

corresponding to a constant curvature spacetime with a vanishing anisotropic stress.

Power-law sector

To exhibit another family of exact solutions for the same matter content, $\rho = \rho_0$, we consider the power-law ansatz

$$\Phi(r) = a\, r^b \,. \tag{20}$$

Let us compute the radial derivatives using spherical coordinates

$$\begin{aligned}\partial_i\partial_j = n_i n_j \partial_r^2 + \frac{1}{r} P_{\perp\, ij}\partial_r + \frac{1}{r}(n_i P_{\perp\, jk} + n_j P_{\perp\, ik})\partial_r\hat{\partial}_k \\ + \frac{1}{r^2}\big(P^k_{\perp\, i} P^l_{\perp\, j}\big)\hat{\partial}_k\hat{\partial}_l \\ - \frac{1}{r^2}\big(n_i P^k_{\perp\, j} + n_j P^k_{\perp\, i} + n_k P_{\perp\, ij}\big)\hat{\partial}_k \,, \end{aligned} \tag{21a}$$

$$\hat{\partial}_k := \frac{\partial}{\partial n^k} \,, \quad P^j_{\perp\, i} := \delta_i\delta^j - n_i n^j \,. \tag{21b}$$

In this case, we have

$$|\boldsymbol{\nabla}\Phi|^2 = a^2 b^2 r^{2b-2} \,, \tag{22a}$$

$$\boldsymbol{\nabla}^2\Phi = ab(b+n-2) r^{b-2} \,, \tag{22b}$$

$$\Phi_{ij}\Phi^{ij} = a^2 b^2 \left[(b-1)^2 + n - 1\right] r^{2b-4}, \tag{22c}$$

$$\Phi_{ij}\Phi^i\Phi^j = a^3 b^3 (b-1) r^{3b-4} \,, \tag{22d}$$

After substituting these into Eq. (78) for the Ricci scalar, we have

$$R = -\frac{b(b+4)(n-1)(n-2)}{4a r^{b+2}} \,. \tag{23}$$

From Eq. (84), the Kretschmann scalar is given by

$$K = \frac{b^2(b+4)^2(n-1)(n-2)}{8a^2 r^{2b+4}} \,. \tag{24}$$

The auxiliary radial function $\mathcal{B}$ is then

$$\mathcal{B} = \frac{(n-2)b(b+4)}{4r^2} \,. \tag{25}$$

For $\rho = 0$, direct integration of Eq. (9) shows that any exponent b is admissible in $n = 2$ spatial dimensions, whereas for $n > 2$ the field equations uniquely select the scalar-flat branch $b = -4$. Eqs. (7c),(8b)&(25) further imply $P = \Pi = 0$. However, as we will see by a change of coordinates, this solution is just Minkowski in disguise, since for $b = -4$ both the Kretschmann (24) and the Ricci (23) scalars vanish. For uniform non-vanishing energy density, $\rho = \rho_0$, the same ansatz instead yields $b = -2$. From Eqs. (23)-(24), both the Ricci and Kretschmann scalars show uniform curvature across the entire spacetime,

$$R = \frac{(n-1)(n-2)}{a} \,, \tag{26a}$$

$$K = \frac{2(n-1)(n-2)}{a^2} \,. \tag{26b}$$

In this solution branch the matter content differs from the Einstein universe, as the Eqs. (7) imply

$$\rho(r) = \frac{\kappa_n^2}{2}\frac{(n-1)(n-2)}{a} \,, \tag{27a}$$

$$P(r) = -\frac{\kappa_n^2}{2}\frac{(n-1)(n-2)^2}{an} \,, \tag{27b}$$

$$\Pi(r) = -\kappa_n^2 \frac{(n-2)}{r^2} \,. \tag{27c}$$

Notably, for $n \geq 3$, generic values of b produce a non-vanishing curvature (Eq. (23)) and hence an effective matter content. Although they are not exact solution branches of the present construction, we include them as a one-parameter geometric deformation for spectral analysis.

Conical structure of the power-law sector

Using the identity (13), the spatial line element becomes

$$\mathrm{d}l^2 = \Phi(r)\left(\mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2_{n-1}\right). \tag{28}$$

The proper radial distance, obtained from the spatial line element in the standard way (see Ref. [16]), is

$$l(r) = \int_0^r \sqrt{\Phi(r')}\,\mathrm{d}r' = \sqrt{a}\int_0^r (r')^{b/2}\,\mathrm{d}r' \,, \tag{29}$$

where the angular coordinates are held fixed, $\mathrm{d}\Omega_{n-1} = 0$.

For $b \neq -2$, this yields

$$l(r) = \frac{\sqrt{a}}{1+\frac{b}{2}}\, r^{1+\frac{b}{2}} \,. \tag{30}$$

At fixed radius, the proper circumference is

$$C(r) = \int_0^{2\pi} \sqrt{\Phi(r)}\, r\, \mathrm{d}\theta = 2\pi\sqrt{a}\, r^{1+\frac{b}{2}} \,. \tag{31}$$

The ratio between circumference and proper radius is therefore

$$\alpha = \frac{C(r)}{2\pi\, l(r)} = 1 + \frac{b}{2} \,, \tag{32}$$

which is independent of r. This shows that the angular sector is uniformly rescaled, corresponding to a conical geometry with deficit (or excess) angle

$$\delta = 2\pi(1-\alpha) = -\pi b \,. \tag{33}$$

This establishes that the spatial geometry associated with the power-law sector is locally conical.

Global monopole spacetime

The conical structure also admits a direct correspondence with the approximate exterior geometry of a global monopole. Indeed, for $b \neq -2$, introducing the radial coordinate

$$\xi := \frac{\sqrt{a}}{|1+b/2|}\, r^{1+b/2} \,, \tag{34}$$

the spatial line element takes the form

$$\mathrm{d}l^2 = \mathrm{d}\xi^2 + \alpha^2\xi^2\mathrm{d}\Omega_{n-1}^2, \qquad \alpha^2 = \left(1+\frac{b}{2}\right)^2. \tag{35}$$

As discussed previously, by using this change of coordinates, the metric $b=-4$ coincides with Minkowski since $\alpha^2=1$.

In three spatial dimensions, the corresponding ultrastatic spacetime coincides with the approximate exterior metric of a point-like global monopole, neglecting the mass contribution associated with its core [5]. In this interpretation, the angular parameter satisfies

$$\alpha^2 = 1 - 8\pi G_{\mathrm{N}}\eta^2, \tag{36}$$

where η denotes the symmetry breaking scale. Consequently, the power-law exponent can be related to the solid-angle deficit through

$$8\pi G_{\mathrm{N}}\eta^2 = 1 - \left(1+\frac{b}{2}\right)^2. \tag{37}$$

This correspondence provides a physical realization of the defect-like geometry arising in the power-law sector and connects its spectral properties with quantum field propagation in global monopole backgrounds.

Cylindrical or Plebański-Hacyan spacetime

The case $b=-2$ is exceptional. In this limit, the spatial line element becomes

$$\mathrm{d}l^2 = a\left(\frac{\mathrm{d}r^2}{r^2} + \mathrm{d}\Omega_{n-1}^2\right). \tag{38}$$

Introducing the logarithmic radial coordinate

$$\zeta := \sqrt{a}\ln\left(\frac{r}{r_0}\right), \tag{39}$$

where r_0 is an arbitrary reference radius, the metric reduces to

$$\mathrm{d}l^2 = \mathrm{d}\zeta^2 + a\,\mathrm{d}\Omega_{n-1}^2. \tag{40}$$

This corresponds to the product geometry $\mathbb{R}\times\mathbb{S}^{n-1}$, where the spherical sections have a constant radius $\sqrt{a}$, independent of the radial coordinate. In particular, the proper circumference remains constant,

$$C = 2\pi\sqrt{a}. \tag{41}$$

Hence, the conical structure is replaced by a generalized cylindrical geometry and the notion of a conical deficit (or excess) angle is no longer applicable.

Including the temporal direction, the full ultrastatic metric takes the form

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}\zeta^2 + a\,\mathrm{d}\Omega_{n-1}^2\,, \tag{42}$$

which describes the direct product $\mathbb{R}^{1+1}\times\mathbb{S}^{n-1}$. In particular, for $(n=3)$, this geometry coincides exactly with the Plebański–Hacyan spacetime, characterized by the product $\mathbb{R}^{1+1}\times\mathbb{S}^2$. Thus, the solution $(b=-2)$ provides a higher-dimensional counterpart of this known spacetime.

Therefore, for $b\neq-2$, the power-law sector describes a defect-like geometry with either a deficit angle $(b<0)$ or an angular excess $(b>0)$. This conical structure coexists with the curvature singularities discussed below and should be understood as a local geometric property away from the singular region. In particular, the presence of a naked curvature singularity for certain parameter values is not necessarily inconsistent with the global monopole interpretation, since the corresponding metric describes only the exterior geometry and does not account for the internal structure of the monopole core. A complete description would therefore require a separate treatment of the core region. This characterization agrees with the standard description of conical geometries, where the deficit angle is determined by the ratio between circumference and proper radius (see, for example, Ref. [17]). From the perspective of quantum field propagation, we expect that this defect-like structure should modify the angular mode decomposition of the field, leading to nontrivial spectral effects.

Geodesic incompleteness and naked singularities

For $n\geq3$, the power-law solutions exhibit curvature singularities whenever $b\neq0,-2,-4$. For $b>-2$, excluding $b=0$, the Ricci scalar (23) diverges as $r\to0$, whereas for $b<-2$, excluding $b=-4$, they diverge as $r\to\infty$. However, in terms of the monopole coordinate ξ introduced in Eq. (34), both limits correspond to $\xi\to0$. Since the geometry depends on b through $\alpha^2=(1+b/2)^2$, the two branches are geometrically equivalent under the transformation $b\mapsto-b-4$. Thus, the curvature singularity is located at $\xi=0$ in both cases. The exceptional values $b=0,-4$ correspond to flat geometries, while $b=-2$ describes a nonsingular cylindrical geometry.

To characterize the singularities, we consider radial time-like geodesics in the ultrastatic background, Eq. (1). The conserved energy $E=\dot{t}$, where the dot denotes differentiation with respect to proper time τ, gives

$$\dot{r}^2 = \frac{E^2-1}{\Phi(r)}\,. \tag{43}$$

Consequently, the proper time required for a radial observer with $E>1$ to reach the singular endpoint from a finite radius r_0 is

$$\Delta\tau = \frac{1}{\sqrt{E^2-1}}\left|\int_{r_0}^{r_{\mathrm{s}}}\sqrt{\Phi(r)}\,dr\right|, \tag{44}$$

where $r_{\mathrm{s}}=0$ for $b>-2$ and $r_{\mathrm{s}}=\infty$ for $b<-2$.

For the power-law configuration $\Phi(r)=r^b$, the integral converges at $r=0$ when $b>-2$ and at $r\to\infty$ when $b<-2$. In fact, using the monopole coordinate ξ,

$$\mathrm{d}\xi = \frac{1+b/2}{|1+b/2|}\sqrt{\Phi(r)}\mathrm{d}r\,, \tag{45}$$

both cases yield

$$\Delta\tau = \frac{\xi(r_0)}{\sqrt{E^2-1}} < \infty\,. \tag{46}$$

Therefore, radial time-like geodesics reach the curvature singularity at $\xi = 0$ in finite proper time, establishing geodesic incompleteness in both curved power-law branches.

Furthermore, the ultrastatic metric satisfies $g_{00} = -1$ throughout the regular region. In particular, radial null signals can propagate from arbitrarily small positive ξ to observers at finite radius without encountering a horizon. The curvature singularities are therefore naked. In contrast, at $b = -2$, the logarithmic Liouville coordinate places $r = 0$ and $r = \infty$ at infinite radial distance, and the resulting cylindrical geometry is geodesically complete.

2 Quantum Correlators

The geometric properties discussed above motivate an investigation of quantum field propagation on these backgrounds. Using canonical quantization in curved spacetime (Ref. [7]) and spectral methods (Refs. [10,18]), we construct the Wightman two-point function for a massless scalar field. We focus on the power-law sector, which includes the approximate exterior geometry of a global monopole and the spectral transition associated with the critical cylindrical configuration at $b = -2$.

Klein-Gordon equation

Let us study the case of a free massless scalar field υ. The Klein-Gordon (KG) equation is given by

$$\ddot{\upsilon} - \frac{1}{\Phi}\boldsymbol{\nabla}^2\upsilon - \frac{n-2}{2\Phi^2}(\boldsymbol{\nabla}\Phi\cdot\boldsymbol{\nabla}\upsilon) = 0\,. \tag{47}$$

The radial symmetry of the background in $n \geq 2$ allows for a separation of variables in Eq. (47). Let us take

$$\upsilon(t,r,\Omega) := T_\omega(t)S_{\ell\omega}(r)Y_{\ell m}(\hat{\mathbf{x}})\,. \tag{48}$$

By the properties of hyperspherical harmonics and the generalized angular momentum operator (See Ref. [19])

$$\hat{L}^2 Y_{\ell m}(\hat{\mathbf{x}}) = \ell(\ell+n-2)\,Y_{\ell m}(\hat{\mathbf{x}})\,, \quad m = \{m_1,\cdots,m_{n-2}\} \tag{49a}$$

$$\hat{L}^2 := -\sum_{i<j} L_{ij}^2\,, \tag{49b}$$

$$L_{ij} := x^i\partial_j - x^j\partial_i\,, \tag{49c}$$

and the generalized Laplacian operator

$$\boldsymbol{\nabla}^2 = \frac{1}{r^{n-1}}\partial_r\big(r^{n-1}\partial_r\big) - \frac{\hat{L}^2}{r^2}\,. \tag{50}$$

Then the ansatz leads to the system of equations

$$\ddot{T}_\omega + \omega^2 T_\omega = 0\,, \tag{51a}$$

$$S''_{\ell\omega} + \left(\frac{n-1}{r} + \frac{n-2}{2}\frac{\Phi'}{\Phi}\right)S'_{\ell\omega} + \left(\omega^2\Phi - \frac{\ell(\ell+n-2)}{r^2}\right)S_{\ell\omega} = 0\,. \tag{51b}$$

In spherical coordinates, we can write the generalized angular momentum operator through the Laplace-Beltrami operator $\hat{L}^2 = -\Delta_{n-1}$. In three dimensions, we have

$$\Delta_2 = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\,. \tag{52}$$

Kernel construction

In order to write the Klein–Gordon radial Eq. (51b) as a proper Sturm–Liouville problem, we introduce the Liouville transformation

$$\mathrm{d}x := \sqrt{\Phi}\,\mathrm{d}r\,, \quad \chi_{\ell\omega}(x) := r^{\frac{n-1}{2}}\Phi^{\frac{n-1}{4}}(r)\,S_{\ell\omega}(r)\,. \tag{53}$$

Then, we can write Eq. (51b) as

$$\left[-\frac{\mathrm{d}^2}{\mathrm{d}x^2} + U(x)\right]\chi_{\ell\omega}(x) = \omega^2\,\chi_{\ell\omega}(x)\,, \tag{54}$$

where the effective Schrödinger potential

$$\begin{aligned} U(x) = \frac{1}{\Phi}\Bigg[&\frac{1}{r^2}\left(\sigma-\frac{1}{2}\right)\left(\sigma+\frac{1}{2}\right) \\ &+ \frac{(n-1)(n-2)}{4r}\frac{\Phi'}{\Phi} + \frac{n-1}{4}\frac{\Phi''}{\Phi} \\ &+ \frac{(n-1)(n-7)}{16}\left(\frac{\Phi'}{\Phi}\right)^2\Bigg], \\ \sigma := \ell - 1 + \frac{n}{2}\,. \end{aligned} \tag{55}$$

implicitly assumes $r = r(x)$. Canonical quantization is then performed on the field χ, while physical observables can subsequently be expressed in terms of the original field υ.

Canonical Quantization

The quantum field $\hat{\chi}$ can be canonically quantized by different representations. Using the spherical symmetry of the background, we expand the canonical field in partial waves

$$\hat{\chi}(t,\mathbf{x}) = \sum_{\ell,m}\hat{\chi}_{\ell m}(t,r)\,Y_{\ell m}(\hat{\mathbf{x}})\,. \tag{56}$$

Following the spectral method of Graham [18] for the massless case, $\omega = k$, the mode expansion for each radial channel consists of a continuum of scattering states and possible discrete bound states

$$\hat{\chi}_{\ell m}(t,\mathbf{x}) = \sqrt{\frac{2}{\pi}}\int_0^\infty \mathrm{d}k \Big(\chi_{\ell\omega}(x)\, T_\omega(t)\, a_{\ell m}(k) + \chi^*_{\ell\omega}(x)\, T^*_\omega(t)\, a^\dagger_{\ell m}(k) \Big) + \sum_j \Big(\chi_{\ell j}(x)\, T_j(t)\, a_{\ell m j} + \chi^*_{\ell j}(x)\, T^*_j(t)\, a^\dagger_{\ell m j} \Big), \quad (57a)$$

$$\big[a_{\ell m}(k), a^\dagger_{\ell' m'}(k')\big] = \delta_{\ell\ell'}\delta_{mm'}\delta(k-k')\,, \quad (57b)$$

$$\big[a_{\ell m j}, a^\dagger_{\ell' m' j'}\big] = \delta_{\ell\ell'}\delta_{mm'}\delta_{jj'}\,, \quad (57c)$$

where the functions $\chi_{\ell j} := \chi_{\ell\omega_j}$ and $\chi_{\ell\omega}$ satisfy the radial channel Eq. (54), while $T_j := T_{\omega_j}$ and T_ω solve the harmonic oscillator, Eq. (51a). Thus, the modes satisfy the usual Wronskian normalization

$$T_\omega(t) = \frac{e^{-i\omega t}}{\sqrt{2\omega}}\,, \quad (58a)$$

$$T_\omega \dot{T}^*_\omega - T^*_\omega \dot{T}_\omega = i\,. \quad (58b)$$

Therefore, the equal-time canonical commutator relation,

$$\Big[\hat{\chi}(t,\mathbf{x}), \hat{\Pi}_\chi(t,\mathbf{x}')\Big] = i\delta(x-x')\delta^{(n-1)}(\hat{\mathbf{x}},\hat{\mathbf{x}}')\,, \quad (59)$$

is satisfied by the completeness relations

$$\sum_{\ell m} Y_{\ell m}(\hat{\mathbf{x}}) Y^*_{\ell m}(\hat{\mathbf{x}}') = \delta^{(n-1)}(\hat{\mathbf{x}},\hat{\mathbf{x}}')\,, \quad (60a)$$

$$\left[\frac{2}{\pi}\int \mathrm{d}k\, \chi_{\ell\omega}(x)\chi^*_{\ell\omega}(x') + \sum_j \chi_{\ell j}(x)\chi^*_{\ell j}(x')\right] = \delta(x-x')\,. \quad (60b)$$

The corresponding radial eigenfunctions satisfy the orthogonality relations

$$\int_0^\infty \chi^*_{\ell\omega}(x)\,\chi_{\ell\omega'}(x)\,\mathrm{d}x = \frac{\pi}{2}\,\delta(\omega-\omega'), \quad (61a)$$

$$\int_0^\infty \chi^*_{\ell j}(x)\,\chi_{\ell j'}(x)\,\mathrm{d}x = \delta_{jj'}\,, \quad (61b)$$

$$\int_0^\infty \chi^*_{\ell\omega}(x)\,\chi_j(x)\,\mathrm{d}x = 0\,. \quad (61c)$$

3 Spectral transition in power-law ultrastatic backgrounds

In this section, we continue the spectral analysis of our ultrastatic backgrounds, focusing on the radial operator and its dependence on the continuous spatial deformation parameter b. Our objective is to characterize the spectral transition between the global monopole geometries and the cylindrical Plebański–Hacyan configuration.

To this end, we consider the power-law metric potential, Eq. (20), with $a = 1$, allowing b to vary across configurations exhibiting curvature singularities, conical profiles, and the nonsingular cylindrical limit.

3.1 Regular radial modes and Dirichlet spectrum for $b > -2$

In the power-law background (20), the radial KG equation (51b) can be written as

$$S''_\ell + \frac{1}{r}\left[(n-1) + \frac{b(n-2)}{2}\right] S'_\ell + \left(\omega^2 r^b - \frac{\ell(\ell+n-2)}{r^2}\right) S_\ell = 0\,. \quad (62)$$

By the change of variable, for $b \neq -2$,

$$S(\rho) = \rho^{\frac{2-n}{2}} Z(\rho)\,, \quad \rho := \frac{2\omega}{b+2} r^{\frac{b+2}{2}}\,, \quad (63)$$

Eq. (62) can be transform into a Bessel equation

$$\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}Z + \frac{1}{\rho}\frac{\mathrm{d}}{\mathrm{d}\rho}Z + \left(1 - \frac{\nu^2}{\rho^2}\right)Z = 0\,, \quad (64a)$$

$$\nu^2(\ell) := \frac{4\ell(\ell+n-2)}{(b+2)^2} + \frac{(n-2)^2}{4}\,. \quad (64b)$$

The general solution is given by a linear combination of Bessel functions or Modified Bessel functions depending of the sign of ω. Since b also changes the background geometry the choice of mode solutions should be studied carefully in accordance with the boundary conditions imposed. In particular, for $b > -2$, only the Bessel function of first kind is regular as

$$r \to 0\,, \quad J_\nu(\rho) \sim \rho^\nu\,.$$

Therefore, the physical solution is given by

$$S_{\ell\omega}(r) = A(\omega)\rho^{\frac{2-n}{2}} J_\nu(\rho)\,. \quad (65)$$

Notably, this is a boundary prescription on the scalar field. However, it does not remove any curvature singularity of the underlying spacetime.

The Liouville coordinate (53) then maps $0 < r \leq L$ to the finite interval $0 < x \leq R_{\mathrm{eff}}(b)$, where

$$R_{\mathrm{eff}}(b) := \int_0^L \sqrt{\Phi(r)}\, dr = \frac{2}{b+2} L^{(b+2)/2}\,. \quad (66)$$

The Dirichlet condition $S_{\ell\omega}(L) = 0$ implies

$$J_{\nu_\ell(b)}\left(\omega_{k\ell} R_{\mathrm{eff}}\right) = 0\,, \quad \omega_{k\ell}(b) = \frac{j_{\nu_\ell(b),k}}{R_{\mathrm{eff}}(b)}\,, \quad k = 1,2,\ldots, \quad (67)$$

where $j_{\nu,k}$ is the kth positive zero of J_ν. Thus, for the regular interior problem with $b > -2$, the radial spectrum is discrete. In particular, $b = 0$ recovers the Minkowski Bessel orders $\nu_\ell(0) = \ell + (n-2)/2$.

In terms of the Sturm–Liouville transformation, Eqs. (53)&(61), the normalized modes are

$$\chi_{\ell\omega}(x) = \sqrt{\rho\frac{\pi}{2}}J_\nu(\rho)\,, \quad x(\rho) = \frac{\rho}{\omega}\,, \quad x(r=0) = 0\,. \tag{68}$$

This coordinate coincides with the radial coordinate (34), $\rho = \omega\xi$, which reveals the monopole structure of the power-law metric.

The case $b = -4$ is geometrically equivalent to the Minkowski limit $b = 0$, as established through the monopole coordinates. In particular, the oscillatory behavior of the Bessel functions as $r \to 0$ corresponds to their large-distance behavior in Minkowski spacetime. The exceptional case $b = -2$, which involves a qualitatively different Sturm–Liouville structure, is treated separately.

3.2 Critical case $b = -2$: a spectral transition

In Sturm–Liouville coordinates (53), the effective radial equation admits a universal scaling form for power-law deformations $\Phi(r) = r^b$, namely

$$U(r) = \frac{c}{r^{b+2}}\,, \tag{69a}$$

$$c := \ell(\ell+n-2) + \frac{(n-1)(n-3)}{16}(b+2)^2\,. \tag{69b}$$

At $b = -2$, the effective Schrödinger equation (54) reduces to a one-dimensional constant potential problem on the half-line in the Liouville coordinate x,

$$\left[-\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \ell(\ell+n-2) - \omega^2\right]\chi_{\ell\omega}(x) = 0\,, \tag{70}$$

and admits solutions of the form

$$\chi_{\ell\omega}(x) = C_1 e^{iqx} + C_2 e^{-iqx}\,, \tag{71a}$$

$$x = \ln\left(\frac{r}{L}\right)\,, \quad 0 < r \le L \quad \longleftrightarrow \quad -\infty < x \le 0\,, \tag{71b}$$

$$q := \sqrt{\omega^2 - \ell(\ell+n-2)}\,. \tag{71c}$$

Then the radial eigenfunctions become plane waves in the logarithmic coordinate. In terms of the radial coordinate r, these solutions correspond to complex power laws, $\chi_{\ell\omega}(r) = C_1 r^{iq} + C_2 r^{-iq}$. In contrast, in the Bessel-type solutions, Eq. (68), for $b > -2$, the radial dependence is governed by a power-law reparametrization.

Geometrically, the full $b = -2$ spacetime is the cylindrical product $\mathbb{R}^{1+1} \times \mathbb{S}^{n-1}$, whose $n = 3$ member is the Plebański–Hacyan geometry. However, the domain $0 < r \le L$ covers only a half-cylinder, with the Dirichlet boundary at $r = L$. Since the logarithmic Liouville coordinate satisfies $x = \ln(r/L) \in (-\infty, 0]$, the limit $r \to 0$ lies at infinite radial distance rather than at a regular center. Consequently, no regular center boundary condition is imposed there.

The transition is already visible from

$$\lim_{b\to-2^+} R_{\text{eff}}(b) = \lim_{b\to-2^+} \frac{2}{b+2}L^{(b+2)/2} = \infty\,. \tag{72}$$

We therefore refer to $b = -2$ as a *spectral critical point*, characterized by the transition from a discrete spectrum on a finite radial interval to a continuous spectrum on a half-line. The $b < -2$ sector maps to an exterior, non-compact radial problem and is not included in the interior mode sum below.

3.3 Physical interpretation

The divergence of the Liouville radial length, the emergence of the half-line problem and its continuous spectrum establish the distinguished nature of $b = -2$. Geometrically, this critical value marks the transition from a conical, global monopole to the Plebański–Hacyan geometry, $\mathbb{R}^{1+1} \times \mathbb{S}^2$, in four spacetime dimensions.

Furthermore, within the global monopole correspondence, the critical value of b is associated with a symmetry-breaking scale approaching the reduced Planck mass, i.e.

$$b \to -2 \Longrightarrow \eta^2 \to M_{\text{P}}^2. \tag{73}$$

However, we interpret this correspondence as signaling the breakdown of the weak-gravity global monopole approximation, rather than establishing a physical transition of a global monopole into a Plebański–Hacyan solution of the full Einstein–matter equations.

4 Final discussion

In this work, we investigated the spectral properties of a massless scalar field in a family of ultrastatic backgrounds characterized by the power-law metric potential $\Phi(r) = a\,r^b$. This family includes solutions of the $(n+1)$-dimensional Einstein field equations and geometries corresponding to global monopole exteriors, characterized by the angular scale factor $\alpha = |1 + b/2|$ for $b \neq -2$. The exceptional case $b = -2$ corresponds to a geometry with constant curvature invariants, uniform energy density, and anisotropic pressure, which can be identified with a higher-dimensional Plebański–Hacyan universe.

Within our framework, we characterized the ultrastatic geometries compatible with the Einstein equations under the prescribed isotropic metric ansatz. The matter sector is described by a radial equation of state relating the pressure and energy density, while anisotropic stress configurations are also admitted. The resulting family connects geometries such as Minkowski spacetime, global monopole exteriors and the Plebański–Hacyan universe through a continuous deformation of the spatial metric.

The main result of this work is the identification of $b = -2$ as a distinguished spectral critical point of the radial massless Klein–Gordon operator. For $b > -2$, the Liouville transformation maps the spherical cavity onto a finite interval, where the radial eigenvalue problem is governed by an inverse-square Schrödinger potential and admits a discrete spectrum of Bessel modes under the adopted boundary conditions. As $b \to -2^+$, the effective radial interval becomes unbounded. At $b = -2$, the Liouville coordinate acquires a logarithmic dependence, $x \sim \ln r$, and the effective Schrödinger potential becomes constant, yielding a continuous spectrum characterized by plane wave solutions. This transition follows from the Sturm–Liouville formulation and is independent of any regularization of the vacuum energy.

The spectral transition is accompanied by a geometric distinction: while the generic power-law configurations exhibit a conical radial structure, the critical geometry becomes a spatial cylinder with topology $\mathbb{R} \times \mathbb{S}^{n-1}$. Thus, the critical configuration is distinguished not only by its regular curvature invariants but also by the qualitative change in the radial spectral problem, from Bessel modes on a finite interval to plane waves on an unbounded domain.

Future work may proceed in several directions. An important extension is the evaluation of the complete spectral zeta function, incorporating the explicit dependence of the Bessel indices on the deformation parameter b, to determine the corresponding renormalized Casimir energy. More generally, the correspondence between the power-law sector and global monopole geometries offers a framework for investigating quantum effects associated with topological defects and their possible cosmological implications.

Supplementary information

No supplementary files are associated with this manuscript.

Statements and Declarations

Funding: No funding was received to assist with the preparation of this manuscript.

Conflict of interest/Competing interests: The authors declare they have no financial interests.

Ethics approval and consent to participate: Not applicable.

Consent for publication: Not applicable.

Data availability: Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Materials availability: Not applicable.

Code availability: Not applicable.

Author contribution: A. F. Cantun: Conceptualization, Methodology, Formal Analysis, Investigation, Writing – original draft, Writing – review & editing.

5 Curvature calculations

The only non–vanishing Christoffel symbols of the metric ansatz (1) are

$$\Gamma^{i}{}_{jk} = \frac{1}{2}\left(\delta^{i}{}_{k}\frac{\partial_j \Phi}{\Phi} + \delta^{i}{}_{j}\frac{\partial_k \Phi}{\Phi} - \delta_{jk}\delta^{il}\frac{\partial_l \Phi}{\Phi}\right). \tag{74}$$

Define

$$h_i := \frac{\partial_i \Phi}{\Phi}, \quad h_{ij} := \partial_i h_j = \frac{\Phi_{ij}}{\Phi} - \frac{\Phi_i}{\Phi}\frac{\Phi_j}{\Phi},$$

then

$$\begin{aligned}
2\big(\partial_k \Gamma^{i}{}_{jl} - \partial_l \Gamma^{i}{}_{jk}\big) &= \delta^i_l h_{kj} + \delta^i_j h_{kl} - \delta_{jl} h^i_k - \delta^i_k h_{lj} - \delta^i_j h_{lk} + \delta_{jk} h^i_l \\
&= \delta^i_l h_{kj} - \delta_{jl} h^i_k - \delta^i_k h_{lj} + \delta_{jk} h^i_l , \qquad (75a) \\
4\Gamma^{i}{}_{km}\,\Gamma^{m}{}_{lj} &= \delta^i_k h_m \delta^m_l h_j + \delta^i_k h_m \delta^m_j h_l - \delta^i_k h_m \delta_{jl} h^m \\
&\quad + \delta^i_m h_k \delta^m_l h_j + \delta^i_m h_k \delta^m_j h_l - \delta^i_m h_k \delta_{jl} h^m \\
&\quad - \delta_{mk} h^i \delta^m_l h_j - \delta_{mk} h^i \delta^m_j h_l + \delta_{mk} h^i \delta_{jl} h^m \\
&= 2\delta^i_k h_j h_l - \delta^i_k \delta_{jl} h^2 + \delta^i_l h_k h_j + \delta^i_j h_k h_l - h^i h_j \delta_{lk} - h^i h_l \delta_{jk} , \qquad (75b) \\
4\big(\Gamma^{i}{}_{km}\,\Gamma^{m}{}_{lj} - \Gamma^{i}{}_{lm}\,\Gamma^{m}{}_{kj}\big) &= 2\delta^i_k h_j h_l - \delta^i_k \delta_{jl} h^2 + \delta^i_l h_k h_j + \delta^i_j h_k h_l - h^i h_j \delta_{lk} - h^i h_l \delta_{jk} \\
&\quad - 2\delta^i_l h_j h_k + \delta^i_l \delta_{jk} h^2 - \delta^i_k h_l h_j - \delta^i_j h_l h_k + h^i h_j \delta_{kl} + h^i h_k \delta_{jl} , \\
&= \delta^i_k (2 h_j h_l - \delta_{jl} h^2) + \delta^i_l h_j h_k - h^i \delta_{jk} h_l \\
&\quad + \delta^i_l (-2 h_j h_k + \delta_{jk} h^2) - \delta^i_k h_j h_l + h^i \delta_{jl} h_k \\
&= \delta^i_k (h_j h_l - \delta_{jl} h^2) + \delta^i_l (-h_j h_k + \delta_{jk} h^2) + h^i (\delta_{jl} h_k - \delta_{jk} h_l) , \qquad (75c)
\end{aligned}$$

Eqs. (75) implicate that the non–vanishing components of the Riemann tensor are

$$\begin{aligned}
R^{i}{}_{jkl} &= \frac{1}{4}\delta^i_l\big(2h_{jk} - h_j h_k + \delta_{jk}h^2\big) \\
&\quad + \frac{1}{4}\delta^i_k\big(-2h_{jl} + h_j h_l - \delta_{jl}h^2\big) \\
&\quad + \frac{1}{4}\delta_{jl}\big(h^i h_k - 2h^i_k\big) + \frac{1}{4}\delta_{jk}\big(-h^i h_l + 2h^i_l\big) \\
&= \delta^i_l\left(\frac{2}{4}\frac{\Phi_{jk}}{\Phi} - \frac{3}{4}\frac{\Phi_j}{\Phi}\frac{\Phi_k}{\Phi} + \frac{1}{4}\delta_{jk}\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^2}\right) \\
&\quad + \delta^i_k\left(-\frac{2}{4}\frac{\Phi_{jl}}{\Phi} + \frac{3}{4}\frac{\Phi_j}{\Phi}\frac{\Phi_l}{\Phi} - \frac{1}{4}\delta_{jl}\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^2}\right) \\
&\quad + \delta_{jl}\left(-\frac{2}{4}\frac{\Phi^i_k}{\Phi} + \frac{3}{4}\frac{\Phi^i}{\Phi}\frac{\Phi^k}{\Phi}\right) + \delta_{jk}\left(\frac{2}{4}\frac{\Phi^i_l}{\Phi} - \frac{3}{4}\frac{\Phi^i}{\Phi}\frac{\Phi^l}{\Phi}\right) \\
&= \frac{1}{2\Phi}\big(\delta^i_l\Phi_{jk} - \delta^i_k\Phi_{jl} - \delta_{jl}\Phi^i_k + \delta_{jk}\Phi^i_l\big) \\
&\quad + \frac{3}{4\Phi^2}\big(-\delta^i_l\Phi_j\Phi_k + \delta^i_k\Phi_j\Phi_l + \delta_{jl}\Phi^i\Phi_k - \delta_{jk}\Phi^i\Phi_l\big) \\
&\quad + \frac{|\boldsymbol{\nabla}\Phi|^2}{4\Phi^2}\big(\delta^i_l\delta_{jk} - \delta^i_k\delta_{jl}\big)\,,
\end{aligned} \tag{76}$$

where $\Phi^i := \delta^{ij}\partial_j\Phi$, $\Phi_j := \partial_j\Phi$ and $h^2 := h^m h_m$.
From this, we can compute the Ricci tensor components as

$$R_{ij} = \left[-\frac{1}{2}\frac{\boldsymbol{\nabla}^2\Phi}{\Phi} + \Big(1 - \frac{n}{4}\Big)\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^2}\right]\delta_{ij} + \Big(1 - \frac{n}{2}\Big)\frac{\partial_i\partial_j\Phi}{\Phi} + \frac{3}{4}(n-2)\frac{\partial_i\Phi\partial_j\Phi}{\Phi^2}\,. \tag{77}$$

The Ricci scalar is then

$$R = (1-n)\frac{\boldsymbol{\nabla}^2\Phi}{\Phi^2} - \frac{(n-1)(n-6)}{4}\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^3}\,. \tag{78}$$

After imposing isotropy on the metric potential, we can express the Ricci spatial components in terms of radial functions,

$$R_{ij} = \mathcal{A}(r)\delta_{ij} + \mathcal{B}(r)\frac{x^i x^j}{r^2}\,, \tag{79a}$$

$$\mathcal{A} := -\frac{1}{2}\frac{\Phi''}{\Phi} + \left(\frac{3}{2} - n\right)\frac{1}{r}\frac{\Phi'}{\Phi} + \frac{1}{2}\Big(2 - \frac{n}{2}\Big)\left(\frac{\Phi'}{\Phi}\right)^2\,, \tag{79b}$$

$$\mathcal{B} := \Big(1 - \frac{n}{2}\Big)\frac{\Phi''}{\Phi} + \Big(\frac{n}{2} - 1\Big)\frac{1}{r}\frac{\Phi'}{\Phi} + \frac{3}{4}(n-2)\left(\frac{\Phi'}{\Phi}\right)^2\,, \tag{79c}$$

$$f' := \frac{\mathrm{d}f}{\mathrm{d}r}\,, \quad r := |\mathbf{x}|\,.$$

Kretschmann scalar

To determine the Kretschmann scalar, we define spatial tensors whose indexes are elevated using exclusively the Kronecker deltas,

$$\mathcal{R}_i{}^{jkl} := \delta_{ii'}\delta^{jj'}\delta^{kk'}\delta^{ll'}R^{i'}{}_{j'k'l'}\,, \tag{80a}$$

$$\omega_{ij} := -\sigma_{ij} + \delta_{ij}h^2\,, \tag{80b}$$

$$\sigma_{ij} := -2h_{ij} + h_i h_j\,, \tag{80c}$$

then we can write the Riemann tensor, Eq. (76), as

$$R^{i}{}_{jkl} = \frac{1}{4}\big(\delta^{i}_{l}\omega_{jk} - \delta^{i}_{k}\omega_{jl} + \delta_{jl}\sigma^{i}_{k} - \delta_{jk}\sigma^{i}_{l}\big)\,. \tag{81}$$

From this, the expression we are seeking is

$$\begin{aligned}
4^2 R^{i}{}_{jkl}\mathcal{R}_{i}{}^{jkl} &= \delta^{i}_{l}\delta^{l}_{i}\omega_{jk}\omega^{jk} - \delta^{i}_{l}\delta^{k}_{i}\omega_{jk}\omega^{jl} + \delta^{i}_{l}\delta^{jl}\omega_{jk}\sigma^{k}_{i} - \delta^{i}_{l}\delta^{jk}\omega_{jk}\sigma^{l}_{i} \\
&\quad - \delta^{i}_{k}\delta^{l}_{i}\omega_{jl}\omega^{jk} + \delta^{i}_{k}\delta^{k}_{i}\omega_{jl}\omega^{jl} - \delta^{i}_{k}\delta^{jl}\omega_{jl}\sigma^{k}_{i} + \delta^{i}_{k}\delta^{jk}\omega_{jl}\sigma^{l}_{i} \\
&\quad + \delta_{jl}\delta^{l}_{i}\sigma^{i}_{k}\omega^{jk} - \delta_{jl}\delta^{k}_{i}\sigma^{i}_{k}\omega^{jl} + \delta_{jl}\delta^{jl}\sigma^{i}_{k}\sigma^{k}_{i} - \delta_{jl}\delta^{jk}\sigma^{i}_{k}\sigma^{l}_{i} \\
&\quad - \delta_{jk}\delta^{l}_{i}\sigma^{i}_{l}\omega^{jk} + \delta_{jk}\delta^{k}_{i}\sigma^{i}_{l}\omega^{jl} - \delta_{jk}\delta^{jl}\sigma^{i}_{l}\sigma^{k}_{i} + \delta_{jk}\delta^{jk}\sigma^{i}_{l}\sigma^{l}_{i} \\
&= n\omega^2 - \omega^2 + \omega_{ij}\sigma^{ij} - \omega\sigma - \omega^2 + n\omega^2 - \omega\sigma + \omega_{ij}\sigma^{ij} \\
&\quad + \sigma_{ij}\omega^{ij} - \sigma\omega + n\sigma^2 - \sigma^2 - \sigma\omega + \sigma_{ij}\omega^{ij} - \sigma^2 + n\sigma^2 \\
&= 2(n-1)\omega_{ij}\omega^{ij} + 4\sigma_{ij}\omega^{ij} - 4\sigma\omega + 2(n-1)\sigma_{ij}\sigma^{ij}\,,
\end{aligned} \tag{82}$$

where

$$\sigma := \sigma^{k}_{k} = h^2 - 2\tilde{h}\,, \quad \tilde{h} := h^{k}_{k}\,, \tag{83a}$$

$$\omega := \omega^{k}_{k} = -\sigma + nh^2\,, \tag{83b}$$

$$\begin{aligned}
\sigma_{ij}\sigma^{ij} &= (h_i h_j - 2h_{ij})\big(h^i h^j - 2h^{ij}\big) \\
&= 4h_{ij}h^{ij} - 4h_{ij}h^i h^j + (h^2)^2
\end{aligned} \tag{83c}$$

$$\begin{aligned}
\omega_{ij}\omega^{ij} &= \big(-\sigma_{ij} + \delta_{ij}h^2\big)\big(-\sigma^{ij} + \delta^{ij}h^2\big) \\
&= \sigma_{ij}\sigma^{ij} - 2\sigma\cdot h^2 + n(h^2)^2\,,
\end{aligned} \tag{83d}$$

$$\sigma_{ij}\omega^{ij} = -\sigma_{ij}\sigma^{ij} + \sigma\cdot h^2\,. \tag{83e}$$

Thus by Eqs. (82)(83) we can write the Kretschmann as

$$\begin{aligned}
K &= g_{ii'}g^{jj'}g^{kk'}g^{ll'}\,R^{i}{}_{jkl}\,R^{i'}{}_{j'k'l'} \\
&= \Phi^{-2}R^{i}{}_{jkl}\mathcal{R}_{i}{}^{jkl} \\
&= \frac{1}{4\Phi^2}\left(\frac{n-1}{2}\omega_{ij}\omega^{ij} + \sigma_{ij}\omega^{ij} - \sigma\omega + \frac{n-1}{2}\sigma_{ij}\sigma^{ij}\right) \\
&= \frac{1}{4\Phi^2}\left[\frac{n-1}{2}\sigma_{ij}\sigma^{ij} - (n-1)\sigma\cdot h^2 + \frac{(n-1)n}{2}(h^2)^2\right. \\
&\qquad\qquad \left. -\sigma_{ij}\sigma^{ij} + \sigma\cdot h^2 + \sigma\cdot\sigma - n\sigma\cdot h^2 + \frac{n-1}{2}\sigma_{ij}\sigma^{ij}\right] \\
&= \frac{1}{4\Phi^2}\left[(n-2)\sigma_{ij}\sigma^{ij} - 2(n-1)\sigma\cdot h^2 + \frac{(n-1)n}{2}(h^2)^2 + \sigma\cdot\sigma\right] \\
&= \frac{1}{\Phi^2}\left[(n-2)h_{ij}(h^{ij} - h^i h^j) + \frac{(n-1)(n-2)}{8}(h^2)^2 + (n-2)\tilde{h}h^2 + \tilde{h}\cdot\tilde{h}\right] \\
&= \frac{1}{\Phi^2}\left[(n-2)\left(\frac{\Phi_{ij}\Phi^{ij}}{\Phi^2} - \frac{3\Phi_{ij}\Phi^i\Phi^j}{\Phi^3} + \frac{2|\boldsymbol{\nabla}\Phi|^4}{\Phi^4}\right) + \frac{(n-1)(n-2)}{8}\frac{|\boldsymbol{\nabla}\Phi|^4}{\Phi^4}\right. \\
&\qquad\qquad \left. + (n-2)\left(\frac{(\boldsymbol{\nabla}^2\Phi)|\boldsymbol{\nabla}\Phi|^2}{\Phi^3} - \frac{|\boldsymbol{\nabla}\Phi|^4}{\Phi^4}\right) + \frac{(\boldsymbol{\nabla}^2\Phi)^2}{\Phi^2} - \frac{2(\boldsymbol{\nabla}^2\Phi)|\boldsymbol{\nabla}\Phi|^2}{\Phi^3} + \frac{|\boldsymbol{\nabla}\Phi|^4}{\Phi^4}\right] \\
&= \frac{(n-2)\Phi_{ij}\Phi^{ij} + (\boldsymbol{\nabla}^2\Phi)^2}{\Phi^4} - \frac{3(n-2)\Phi_{ij}\Phi^i\Phi^j}{\Phi^5} + \frac{(n-4)(\boldsymbol{\nabla}^2\Phi)|\boldsymbol{\nabla}\Phi|^2}{\Phi^5} \\
&\qquad\qquad + \frac{(n-1)(n+6)}{8}\frac{|\boldsymbol{\nabla}\Phi|^4}{\Phi^6}\,.
\end{aligned} \tag{84}$$

References

1. L.H. Ford, *Quantum vacuum energy in general relativity*, *Phys. Rev. D* **11** (1975) 3370.
2. J.S. Dowker and R. Critchley, *Covariant casimir calculations*, *Journal of Physics A: Mathematical and General* **9** (1976) 535.
3. A.A. Popov, *Analytical approximation for φ^2 of a quantized scalar field in ultrastatic asymptotically flat spacetimes*, *Phys. Rev. D* **70** (2004) 084047.
4. B.F. Svaiter and N.F. Svaiter, *Zero point energy and analytic regularizations*, *Phys. Rev. D* **47** (1993) 4581.
5. M. Barriola and A. Vilenkin, *Gravitational field of a global monopole*, *Phys. Rev. Lett.* **63** (1989) 341.
6. V.S. Barroso and J.P.M. Pitelli, *Vacuum fluctuations and boundary conditions in a global monopole*, *Phys. Rev. D* **98** (2018) 065009.
7. N.D. Birrell and P.C.W. Davies, *Quantum Fields in Curved Space*, Cambridge University Press, Cambridge (1982).
8. S.W. Hawking, *Zeta function regularization of path integrals in curved spacetime*, *Communications in Mathematical Physics* **55** (1977) 133.
9. E. Elizalde, *Ten physical applications of spectral zeta functions*, vol. 35 of *Lecture Notes in Physics*, Springer-Verlag, Berlin (1995), 10.1007/978-3-540-44757-3.
10. S.A. Fulling, *Vacuum energy as spectral geometry*, *Symmetry, Integrability and Geometry: Methods and Applications* **3** (2007) 094.
11. S.A. Fulling, C.S. Trendafilova, P.N. Truong and J. Wagner, *Wedges, cones, cosmic strings and their vacuum energy*, *Journal of Physics A: Mathematical and Theoretical* **45** (2012) 374018.
12. A.F. Cantun, *Classical and semiclassical probes of dimension-dependent effective geometries*, *The European Physical Journal Plus* **141** (2026) 1136.
13. J.F. Plebański and S. Hacyan, *Some exceptional electrovac type d metrics with cosmological constant*, *Journal of Mathematical Physics* **20** (1979) 1004.
14. D. Gajic, *Linear waves on constant radius limits of cosmological black hole spacetimes*, *Advances in Theoretical and Mathematical Physics* **22** (2018) 919 [1412.5190].
15. R. Emparan and H.S. Reall, *Black holes in higher dimensions*, *Living Reviews in Relativity* **11** (2008) 6.
16. R.M. Wald, *General Relativity*, Chicago Univ. Pr., Chicago, USA (1984), 10.7208/chicago/9780226870373.001.0001.
17. A. Vilenkin, *Cosmic strings*, *Phys. Rev. D* **24** (1981) 2082.
18. N. Graham, R. Jaffe, V. Khemani, M. Quandt, M. Scandurra and H. Weigel, *Calculating vacuum energies in renormalizable quantum field theories:: A new approach to the casimir problem*, *Nuclear Physics B* **645** (2002) 49.
19. Z.-Y. Wen and J. Avery, *Some properties of hyperspherical harmonics*, *Journal of Mathematical Physics* **26** (1985) 396.
20. S. Leseduarte and A. Romeo, *Complete zeta-function approach to the electromagnetic casimir effect for spheres and circles*, *Annals of Physics* **250** (1996) 448.